

GEEM 433

ENGINEERING ETHICS AND MORALE
PHILOSOPHY



Marks Distribution at a glance

- Total Available Points:
- Class Tests:
- Midterm Exam:
- Final Exam:
- Case Studies:
- Group Presentation:



What will be covered in this course

Course Content

- Engineering Ethics:** Introduction to Ethics; Theories of Ethics; Principles of Engineering Ethics; Ethical expectation: Employers and employees, Inter-professional relationship,
- Standards and codes:** Fundamental Canons, NSPE (National Society of Professional Engineers) codes, IEEE codes of conduct, ACM codes; Institutionalization of ethical conduct. Ethical Dilemmas, Choices (Whistleblowing),
- Computer Ethics:** Computer Crime and Cyber Security, Privacy and Confidentiality issues in CSE, Legal Framework in CSE-Copyright laws, ICT Act, Right To Information (RTI), Patents, and Royalty etc. Ethical Challenges for CSE Engineers with the Advancement of Technology;
- Case studies** related to ethical issues in ICT and other Engineering disciplines.
- Introduction to **Philosophy of Engineering**, metaphysics, epistemology, axiology, and logic.

What will be covered in this course

- This course motivates engineers to perform under a standard of **professional behavior** that requires compliance to the **highest principles of ethical conduct** and **managing resources and decisions** effectively.
 - Part of professional ethics is the understanding of the ethics of other professions: how they **interact** and what can be **expected** from them as correct ethical behavior. It **improves** the profession, helps **set better rules** for the future, and **shapes** the way people think and act ethically.

Scenario:

- An engineering firm is hired to design a smart **healthcare monitoring system** in partnership with a hospital. The engineers must work closely with medical professionals and data privacy officers.
- Professional Behavior & Ethical Conduct?
- Outcome?

What will be covered in this course

Scenario:

- An engineering firm is hired to design a smart healthcare monitoring system in partnership with a hospital. The engineers must work closely with medical professionals and data privacy officers.
- Professional Behavior & Ethical Conduct:
 - The engineers ensure patient data is encrypted
 - They respect the doctors' ethical obligation
 - Avoid unethical vendors
- Outcome:

By understanding and respecting the ethics of the **medical profession**, the engineers contribute to a solution that is **technically sound, ethically responsible, and socially beneficial**.



What will be covered in this course

- **Objectives**

1. To develop a firm ethical base.
2. Understanding Legal Issues & Professional Development
3. Assessing Professional Conduct Codes
4. To identify and analyze practical legal problems commonly encountered in the computing industry.

What Does Ethics Mean?

- Ethics refers to the principles and values guiding individuals or groups in determining right or wrong, good or bad, just or unjust, virtue and vice. It helps people decide how to act in various situations, emphasizing fairness, respect, and responsibility.



Student Code of Ethics

❑ Plagiarism

Using someone else's words, code, or ideas **without credit**.

❑ Cheating

Using **unauthorized help or resources** during exams or assignments.

❑ Copying

Submitting work that was **copied from a classmate** or letting someone else copy yours.

❑ Multiple Submission

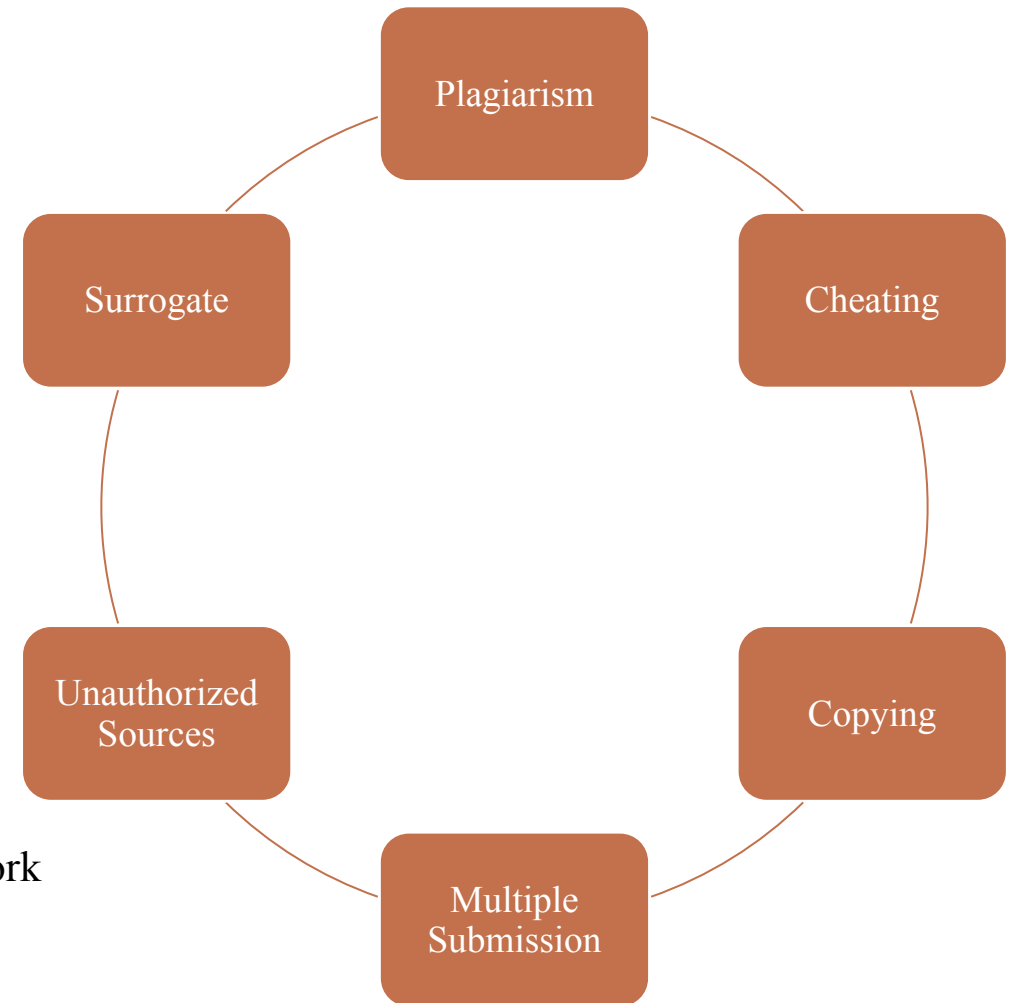
Submitting the **same work for more than one course or assignment** without permission.

❑ Unauthorized Sources

Using **external services or people** to complete your academic work without permission.

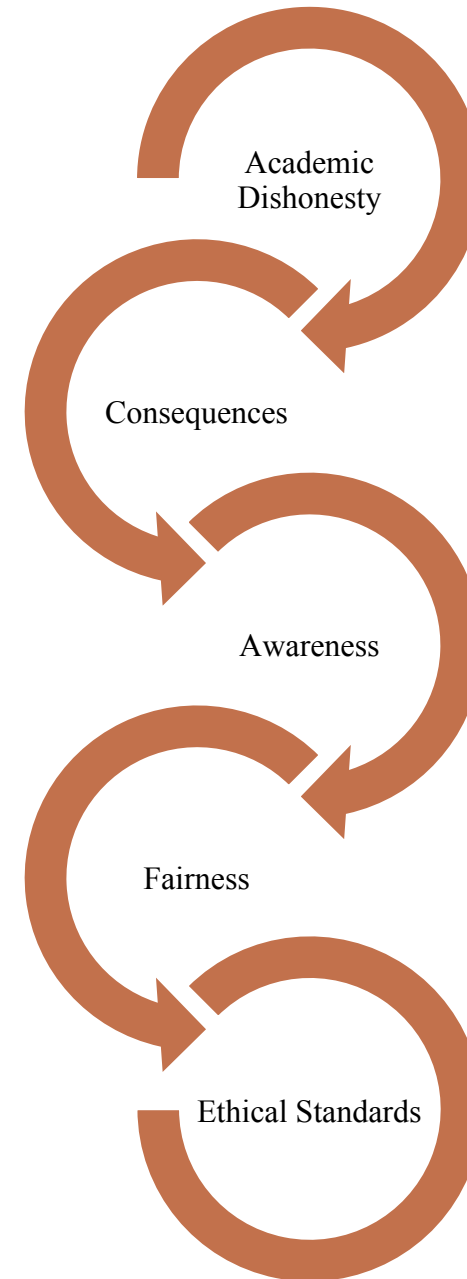
❑ Surrogate

Having **someone else complete work or attend exams** on your behalf.



Student Code of Ethics

- What should we do to maintain the highest Ethical Standards?
- **Avoid Academic Dishonesty**
Don't cheat, plagiarize, or use surrogates. Always submit your own original work.
- **Understand the Consequences**
Violations can lead to failing grades, disciplinary action, or expulsion.
- **Raise Awareness**
Know what counts as unethical behavior (e.g., copying code, unauthorized collaboration).
- **Practice Fairness**
Treat others' work and efforts with respect. Don't gain an unfair advantage over classmates.
- **Uphold Ethical Standards**
Follow your institution's code of conduct. Make ethical decisions even when no one is watching.



Ethics

- **Why do we need it?**
 - Ensure harmony of the group while providing mutual benefits to the individual
 - Basis for how we should treat others
 - It overrides self-interests

Moral Reasoning

- Moral reasoning is the process of **thinking** carefully about what is right or wrong, fair or unfair, and good or bad in a situation — especially when it affects other people.
- It's a way of **making decisions based on values** like:
 - ✓ **Justice** (Is it fair?)
 - ✓ **Equality** (Is everyone treated the same?)
 - ✓ **Freedom** (Are people free to choose?)
 - ✓ **Health and Safety** (Will this harm or help someone?)

নৈতিক যুক্তি

Moral Reasoning

Why It's Important?

- Moral reasoning helps us:
 - ✓ Make **ethical choices**
 - ✓ Stand up for **what's right**
 - ✓ Understand **consequences**
 - ✓ Become **responsible citizens or professionals**

Social Responsibility

- **Social Responsibility** requires taking into consideration the **needs of society**.
- **Engineers** must consider designing, building, and marketing products that benefit **society**.
- **One main connection** between ethics and engineering comes from how **engineering products** and **processes** affect society and their impacts.
- Ethics is about doing what's **right**, and engineering affects society—so ethical engineers must care about **people, safety, fairness, and the environment**.

Social Responsibility

- **Thinking Beyond the Project**

Engineers must ask:

- "Will this product be safe?"
- "Is it environmentally friendly?"
- "Does it help people or harm them?"

- **Examples of Social Responsibility in Engineering:**

- **Designing energy-efficient buildings** to reduce environmental impact.
- **Creating affordable medical devices** so more people can get treatment.
- **Developing apps** that protect user privacy and data.

Ethics in Engineering

Ethics is part of engineering for two main reasons.

- a) Engineers need to be socially responsible when building products and processes for society.**
- b) Social responsibility requires professional responsibility through obligations and duties.**

Typical Ethical Issues Encountered

- Safety
- Acceptable risk
- Compliance
- Confidentiality
- Environmental health
- Data integrity
- Conflict of interest
- Honesty/Dishonesty
- Societal impact
- Fairness
- Accounting for uncertainty, etc.

Typical Ethical Issues Encountered

- **Safety**

Ensuring systems or products do not harm users.

→ *Example:* An engineer refuses to approve a bridge design that doesn't meet load requirements.

- **Acceptable Risk**

Balancing innovation with the risk it brings.

→ *Example:* Launching a self-driving car only after extensive safety testing.

- **Compliance**

Following laws, regulations, and standards.

→ *Example:* A factory follows environmental laws to limit emissions.

- **Confidentiality**

Protecting sensitive or private information.

→ *Example:* A software developer keeps user data secure and does not share it with advertisers.

Typical Ethical Issues Encountered

- **Environmental Health**

Considering the impact of engineering activities on the environment.

→ *Example:* Using eco-friendly materials in construction projects.

- **Data Integrity**

Ensuring data is accurate, secure, and not manipulated.

→ *Example:* A data analyst refuses to falsify research results under pressure.

- **Conflict of Interest**

Avoiding situations where personal gain affects professional judgment.

→ *Example:* A government engineer steps aside from approving a project submitted by their brother's company to avoid a conflict of interest.

- **Honesty/Dishonesty**

Being truthful in reporting, design, testing, and communication.

→ *Example:* Reporting flaws in a software system instead of hiding them.

Typical Ethical Issues Encountered

- **Societal Impact**

Thinking about how a project affects society.

→ *Example:* Developing affordable water purification systems to improve public health in rural communities.

- **Fairness**

Ensuring equal treatment and access for all.

→ *Example:* Developing AI algorithms that are fair to all demographic groups.

- **Accounting for Uncertainty**

Being transparent about unknowns, risks, or limitations.

→ *Example:* Clearly communicating the limits of a predictive model to users.

Code of Ethics – Some thoughts!

- **It is not a legally binding document.**
- **It is not something that engineers need to memorize.**
- **It is something engineers should understand and live by in their works**
- **However, in the beginning knowing the code is a guide to understanding how to apply it.**